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Global Technology

AI Supply Chain: Server Racks Catching Up, What's Next?

NVL72 server rack ramp is a major relief to global AI supply chain, but "1k per week" remains a target to meet. AI GPU shipments to China are another key development to watch.

Key Takeaways

- Joe Moore reiterated NVIDIA as Top Pick on strong Blackwell demand. Charlie Chan reiterated TSMC as Top Pick on intact CoWoS-L order and wafer price hike.
- Howard Kao switched Top Pick to Gigabyte on NVL72 exposure.

NVL72 server rack ramp the key focus: We have been using [Exhibit 2](#) to show the gap between TSMC's Blackwell chip output and downstream OEM's NVL72 server rack shipments. The chart assumed 30k of NVL72 server racks in 2025. After Computex, our tech downstream team analysts Sharon Shih and Howard Kao raised their assumptions to 25-30k ([link](#)). In [NVIDIA's](#) (covered by Joe Moore) recent earnings call, the company commented that "**multiple customers are taking 1k rack per week**", Joe thinks it seems to be more a pipeline comment and may include some HGX racks. We have been hearing that after major issues in January that "the rack solutions were identified", and Joe has been saying that the rack numbers will start being revised higher – and it seems that's now the case. On TSMC, we therefore feel more comfortable about our Street-high 390k CoWoS-L (for Blackwell chips) in 2025 vs. Taiwanese supply-chain chatter of a cut to 340k.

What's next key events/data points to watch from Asia supply chain?

- 1) Taiwan monthly sales** and its implied shipments of NVL72 server racks from Taiwan downstream ODMs Hon Hai, Quanta and Wistron. Total 3-3.5k per month is the step to reach 10k+ per quarter (or "1k per week").
- 2) TSMC 2026 CoWoS forecast:** Of note, we are not bearish on 2026 demand; we do not think TSMC will have a concrete forecast on 2026 CoWoS consumption until this July, and thus keep our 2026e CoWoS at 90k (vs. Street's 115k). Our current assumptions still imply 31% Y/Y growth in cloud AI semis in 2026, which should be in line with the AI capex growth, if we assume the AI server mix to grow from 50% in 2025, to 65% in 2026, at the major CSPs (see [Exhibit 1](#)).
- 3) China version GPU demand and supply:** As of NVIDIA's earning call, the new license for shipping AI GPU to China had not been granted. Our supply-chain checks suggest the [B30 China version GPU design](#) could be a similar to the RTX PRO 6000 that was showcased during Computex (see [Exhibit 3](#) and [Exhibit 4](#)). Before Computex, we saw 0.5mn units volume of B30 (or RTX PRO 6000), but demand continues to grow. In [Digitimes'](#) recent news report, it reported AMD may introduce a gaming-graphic-based GPU to replace MI308 GPU to ship to China.

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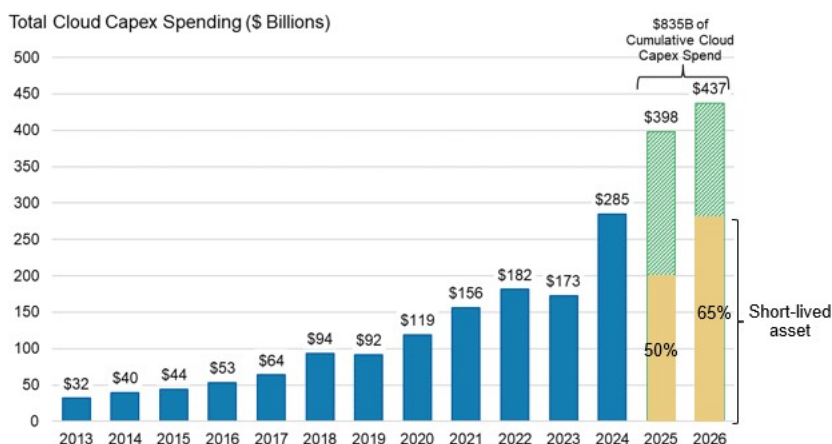
Key charts that address the AI hardware demand and supply debates

From the recent earnings calls of global cloud CSP, a 50% share of the US\$398bn in total capex should be a good gauge of the AI server TAM for 2025 (around US\$199bn).

We at Morgan Stanley Research have been forecasting 10% Y/Y growth of global cloud capex in 2026 (see [Exhibit 1](#)). If we assume the AI server mix grows from 50% in 2025 to 65% in 2026, at the major CSP, global AI capex growth in 2026 could be higher than 30% Y/Y.

Although the quarterly growth momentum could be flat in 2026, the implied full-year demand profile is still sizable for AI hardware.

Exhibit 1: The mix of AI servers in Global Cloud Capex is an important demand indicator for AI hardware



Source: Company data, Morgan Stanley Research (e) estimates; note: estimates are compiled using our supply chain checks

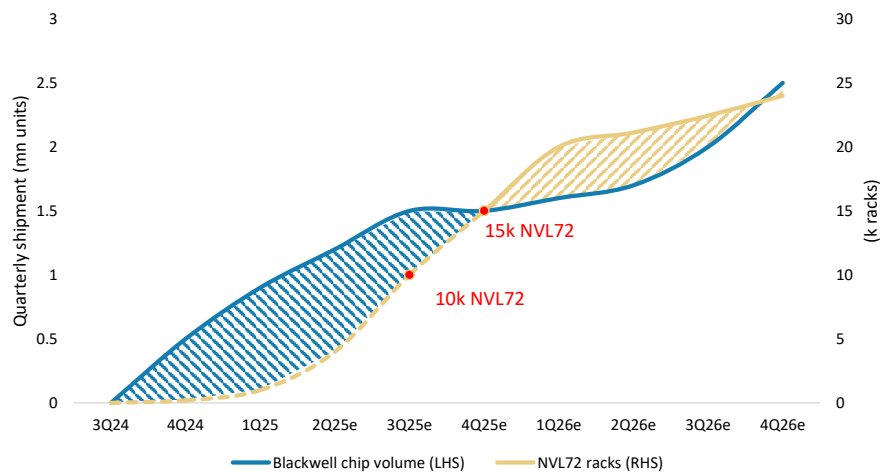
We think TSMC's Blackwell chip outputs (calculated by CoWoS-L capacity) aligns more to the strong "demand" (which is measured by the above-mentioned AI capex). However, in the first few quarters of the Blackwell product cycle (4Q24 to 3Q25), chip outputs should exceed the NVL72 server racks assembly, and that could generate some chip inventory (the blue area in [Exhibit 2](#)).

We have been hearing that after major issues in January, "the rack solutions were identified", and Joe has been saying that the rack numbers will start being raised again, which seems to be the case now. On NVIDIA's (covered by Joe Moore) recent earning call, the company commented that "multiple customers are taking 1k rack per week". This dynamic may translate into chip inventory accumulation (blue area in [Exhibit 2](#)) to the chip inventory digestion (the yellow area) in 1H26.

On TSMC, we therefore feel more comfortable about our Street-high 390k CoWoS-L estimate (for Blackwell chips) in 2025 vs. Taiwanese supply-chain chatter of a cut to 340k.

Our current assumption implies 31% Y/Y of cloud AI semis in 2026, which should be in line with the AI capex growth outlook. Of note, we are not bearish on 2026 demand; it is just that we do not believe TSMC will have a concrete forecast on 2026 CoWoS consumption until this July. We are assuming downstream OEMs digest chip output in 1H26, and keep our 2026 year-end CoWoS estimate at 90kwpm (vs. market's 115kwpm).

Exhibit 2: TSMC is expected to produce 5.1mn of chips in 2025, with full-year GB200 NVL72 shipments expected to reach 30k



Source: Company data, Morgan Stanley Research (e) estimates. Note: Estimates are compiled using our supply chain checks and may differ from other MS estimates.

Updates and Introduction on NVIDIA's new RTX6000 PRO product

We think RTX PRO 6000 GPU server could be the new demand driver for the China market. NVIDIA announced its new RTX PRO 6000 GPU server during Computex 2025. This server could have eight Blackwell GPUs on one computing tray with GDDR7 memory. NVIDIA also announced its latest CX8 (ConnectX-8) SuperNIC switch to interconnect GPUs with NVLink and CPU/GDDR7 with PCIe. We saw some ODMs showcase this RTX PRO 6000 MGX server (modular reference design) at Computex booths that could have four computing trays in a server rack. The interconnect on CX8 could be as fast as PCIe Gen 6, while the connect to memory module could be PCIe Gen 5. We believe this server should be used for inferencing, given the bandwidth limitation of GDDR. However, we also believe this could not only fulfil the demand of US enterprise but also strong inferencing demand from China as H2O with HBM is not currently allowed for China market.

In our [Computex preview](#) report, we saw 0.5mn units volume of B30 (or RTX PRO 6000), but the demand continues to grow. In [Digitimes'](#) recent news report, it reported that AMD may introduced a gaming-graphic-based GPU to replace MI308 AI GPU to ship to China.

Exhibit 3: RTX Pro 6000 Blackwell GPU



Source: Foxconn, Morgan Stanley Research

Exhibit 4: ConnectX-8 with Built-in PCIe Gen6 Switch



Source: Foxconn, Morgan Stanley Research

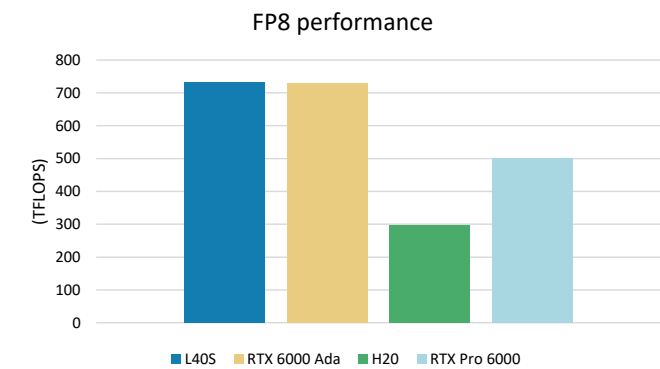
Exhibit 5:

Comparison between L40S, RTX 6000 series and H20

Product	Nvidia			
	L40S	RTX 6000 Ada	H20	RTX Pro 6000
Node	TSMC N4	TSMC N4	TSMC N4	TSMC N4
FP16 (TFLOPS)	362	364	148	250
FP8/Int8 (TFLOPS)	733	729	296	500
FP4 (TFLOPS)	NA	NA	NA	1,001
HBM usage	GDDR6	GDDR6	HBM3	GDDR7
HBM memory (GB)	48	24	96	96
HBM bandwidth (GB/s)	864	672	4,000	1,792
GPU-GPU networking bandwidth	PCIe Gen4: 64GB/s	NVLink: 100GB/s	NVLink: 900GB/s	PCIe Gen5: 128GB/s
TDP (W)	350w	300W	400W	600W

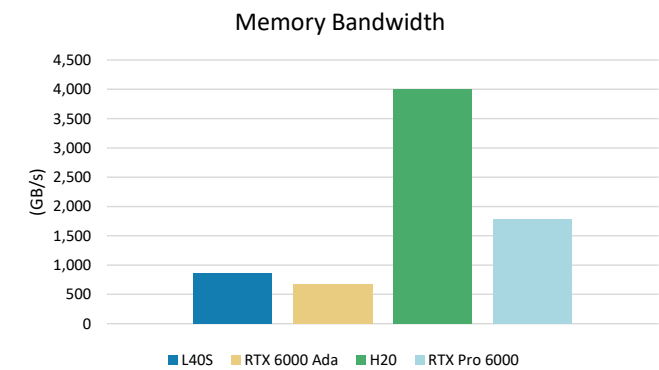
Source: NVIDIA, Morgan Stanley Research

Exhibit 6: L40S and RTX 6000 series products all have better FP8 performance than H20



Source: NVIDIA, Morgan Stanley Research

Exhibit 7: While H20 possesses greater memory bandwidth



Source: NVIDIA, Morgan Stanley Research

AI ASIC design activity

Key debate on AWS ASIC projects

- AWS Trainium 2: On current Trainium2 volume, in Asia we pick up different numbers from upstream and downstream. We are seeing 1mn units on this upstream side, but our downstream analyst Derrick Yang's checks indicate that there could be over 1.5mn units in 2025 (with shipments extending into 2H26). For 2026, we assume T2 ramps down, as T3 volume could pick up to 500k units.
- AWS Trainium 3:
 - At its recent earnings call, Marvell (covered by Joe Moore) management was still confident in next-generation Trainium3 at AWS, although acknowledging there could be some uncertainties. Joe thinks the AWS Trainium3 starting later this year remains a debate. While Joe thinks there can be some partial loss of share to competitor Alchip, he is also hearing conviction from multiple sources within Marvell – including but not limited to management – that it does have major ASIC projects for 2026. Please see [more details from Joe's Marvell report](#).
 - We are confident on Alchip's 3nm project market share and production timing; TSMC's CoWoS capacity allocation will soon prove it, we think. According to our foundry industry checks, Alchip taped out the 3nm ASIC in February, and that design recently saw wafer outputs from the TSMC foundry. Therefore, we believe Alchip will be on track to enter small production of 3nm by 2025, and then a bigger volume production from early 2026. We estimate 30k wafers of CoWoS to produce 500k Trainium3 chip, worth US\$1.5bn revenue for Alchip in 2026.
- AWS Trainium 4: Our Asia team also thinks it is highly likely that Alchip can continue to win 2nm (given lower cost, better service and third-party IP support), and that an early decision could be made by AWS in 3Q25.

Exhibit 8: ASIC mapping table

	Alchip	Global Unichip	Broadcomm	Marvell	TSMC direct	Samsung direct	MediaTek	Socionext	Intel direct
AWS – Annapurna	Inferentia2/Trainium1 (7nm) Trainium3 (3nm)			Trainium2/Trainium2.5 (5nm)	Graviton 1 (16nm) Graviton 2 (7nm) Graviton 3 Nitro Networking Chip				AI fabric chips (18A) Xeon 6 (3nm)
Amazon – Lab 126	Kindle processor Echo processor (12nm)	Echo processor (5nm)							
Google		Security Chip Axion2 CPU (3nm) Axion3 CPU (IP Only)	TPU 1/TPU 2 TPU 3/TPU 4 TPU 5/TPU 6e (Trillium) TPU 7p (Ironwood; 5nm) TPU 7e (5nm)	Maple CPU (5nm) Axion3 CPU (7nm)	Axion1 CPU (5nm) Tensor (3nm)	Tensor (7nm) Tensor (5nm)	TPU 3 TPU 8p (3nm) TPU 8e (3nm)		
Microsoft		Maia 100 (5nm) Cobalt 100 (5nm) Maia 200 (3nm)		Maia 200 enhancement (3nm)			Azure IoT XBox Bluray		
Meta	Oculus ASIC		MTIA v1 MTIA v2 MTIA v3	Switch Networking AI Networking			Oculus ASIC	CPU (3nm)?	
Apple		MTIA v3 (CoWoS)		Custom Networking Chip					
Sony	TV ASIC Smartphone ISP	DSC ASIC					PSS South Bridge		
Tesla/xAI	D1/Dojo (7nm)	AI accelerator			D1/Dojo (7nm) D2/Dojo (5nm)	Autopilot 3.0/FSD 1 (14nm) Autopilot 4.0/FSD 2 (7nm)			
Li Auto	ADAS High-end (5nm) ADAS Low-end (5nm)								
GM – Cruise								5nm	
Intel – Habana	Gaudi 1 (16nm) Gaudi 2/Goya 2 (7nm) Gaudi 3/Goya 3 (5nm)	Goya 1 (16nm)							
OpenAI			AI Custom Chip						
Baidu					Kunlun 2 (7nm)	Kunlun 1 (14nm) Kunlun 3 (5nm)			
Alibaba – T-head		Generation 1			Hanguan				
Tencent					Zixiao (12nm)				
ByteDance			AI accelerator (7nm) 5nm?				AI accelerator (3nm)		

Source: Company data, Morgan Stanley Research

CoWoS allocation assumptions

Exhibit 9: Global CoWoS demand breakdown

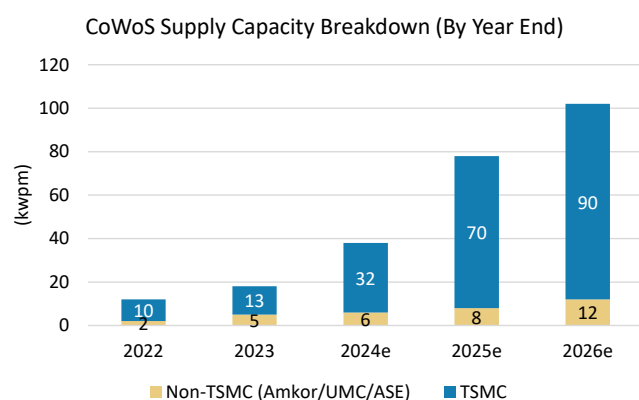
(k wafer)	2023	2024e	2025e	2026e	2022e	2023e	2024e	2025e	2026e
NVIDIA	53	200	425	580	40%	45%	54%	64%	68%
CoWoS-L			390	540				58%	63%
CoWoS-S			20	10				3%	1%
CoWoS-R			15	30				2%	4%
Broadcom	23	68	85	100	25%	20%	18%	13%	12%
AMD	7	40	50	59	2%	6%	11%	7%	7%
Xilinx	3	10	10	10	3%	3%	3%	1%	1%
MediaTek				25					3%
AWS + Alchip	9	16	5	30	1%	8%	4%	1%	4%
Intel Habana	0	7	9	2	0%	0%	2%	1%	0%
Marvell	1	18	65	25	3%	1%	5%	10%	3%
GUC	1	1	1	1	0%	1%	0%	0%	0%
Others	20	10	19	25	27%	17%	3%	3%	3%
Total demand	117	370	669	857	100%	100%	100%	100%	100%

Source: Company data, Morgan Stanley Research (e) estimates; note: estimates are compiled using our supply chain checks

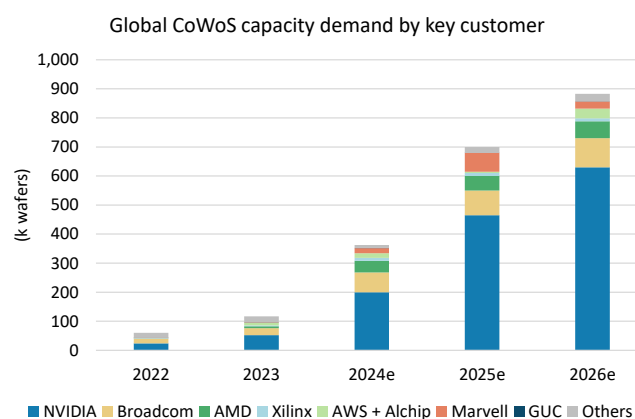
Exhibit 10: Global CoWoS demand Y/Y growth profile

Y/Y	2023	2024e	2025e	2026e
NVIDIA	119%	280%	113%	36%
Broadcom	56%	191%	25%	18%
AMD	485%	470%	25%	17%
Xilinx	63%	242%	0%	0%
MediaTek				
AWS + Alchip		71%	(69%)	500%
Intel Habana			29%	(78%)
Marvell	(22%)	1438%	261%	(62%)
GUC		(15%)	100%	0%
Others	23%	(49%)	90%	32%
Total demand	95%	216%	196%	31%

Source: Company data, Morgan Stanley Research (e) estimates; note: estimates are compiled using our supply chain checks

Exhibit 11: Global CoWoS capacity expansion by year end and by vendor

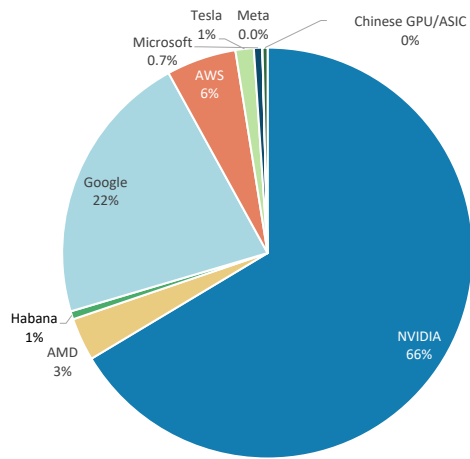
Source: Company data, Morgan Stanley Research (e) estimates

Exhibit 12: Global CoWoS consumption, by customer

Source: Company data, Morgan Stanley Research (e) estimates; note: estimates are compiled using our supply chain checks

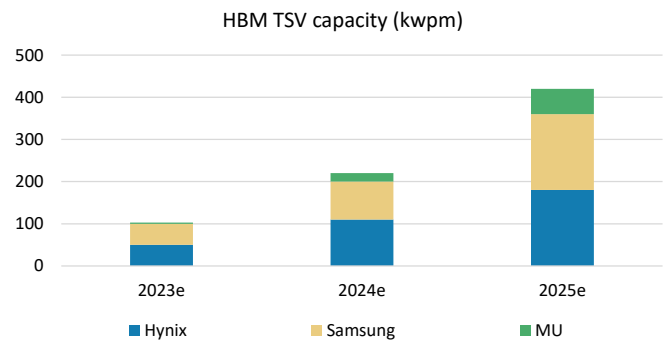
2025 AI semi foundry wafer revenue and HBM demand calculation

Exhibit 13: Leading-edge wafer consumption, by customer, 2025e



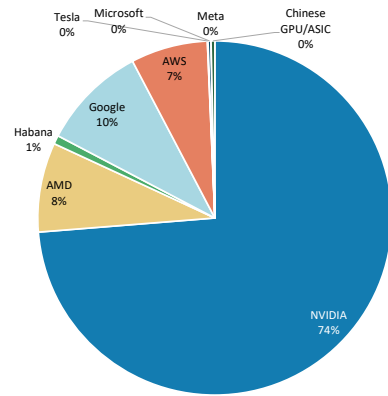
Source: Company data, Morgan Stanley Research (e) estimates; note: estimates are compiled using our supply chain checks

Exhibit 15: HBM TSV capacity appears set to double in 2025



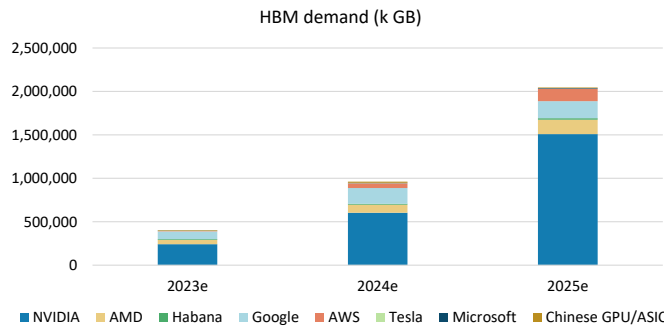
Source: TrendForce, Morgan Stanley Research (e) estimates

Exhibit 14: HBM consumption in 2025e – NVIDIA seen as largest customer



Source: Company data, Morgan Stanley Research (e) estimates; note: estimates are compiled using our supply chain checks

Exhibit 16: We expect 2025 HBM demand to be nearly double 2024 level



Source: TrendForce, Morgan Stanley Research (e) estimates

Exhibit 17:

AI computing wafer consumption in 2025e – up to US\$14.8bn revenue

AI chip vendor	Product name	CoWoS capacity allocation (k wafers)	Chips per CoWoS wafer	Implied shipments (k)	Compute die size	Geometry	Compute die units	Wafer consumption (k wafers)	Wafer price (US\$)	Wafer revenue TAM (US\$ mn)
AI GPU (2025e)										
NVIDIA	H200/H20	50	27	1,350	814	4nm	1	38	20,900	789
	B200	160	14	2,240	850	4nm	2	178	20,900	3,716
	B300A	28	27	745	850	4nm	1	30	20,900	618
	B300	202	14	2,834	850	4nm	2	225	20,900	4,700
AMD	MI300	5	12	60	110	5nm	8	2	25,200	44
	MI325	20	12	240	110	5nm	8	7	25,200	176
	MI355	25	12	300	110	3nm	8	9	32,004	279
Habana	Gaudi 3	9	14	126	750	5nm	2	5	20,000	95
AI ASIC (2025e)										
Google	TPU v5 training (v5p)	10	25	250	600	5nm	1	6	25,200	143
	TPU v6 inferencing (v6e)	60	30	1,800	600	3nm	2	82	32,004	2,613
	TPU v6 training (v6p)	15	20	300	600	3nm	2	14	32,004	436
AWS	Trainium 2	40	16	640	600	5nm	2	24	22,000	521
	Trainium 2.5	25	16	400	600	5nm	2	11	22,000	251
	Trainium 3	5	17	85	700	3nm	1	1	27,940	41
Tesla	Dojo 1	NA	NA	200	645	7nm	1	3	15,400	54
	FSD	NA	NA	1,600	300	7nm	1	10	15,400	160
Microsoft	Maia 200	2	29	58	700	3nm	1	1.5	26,400	41
	Maia 200 Enhancement	1	29	29	700	3nm	1	0.8	26,400	20
Chinese GPU/ASIC	Chinese GPU/ASIC	5.5	20	110	1,000	7nm	1	3	18,700	59
Total		663						651		14,793

Source: Company data, Morgan Stanley Research (e) estimates; note: estimates are compiled using our supply chain checks

Exhibit 18: HBM consumption in 2025e – up to 18bn GB

AI chip vendor	Product name	CoWoS capacity allocation (k wafers)	Chips per CoWoS wafer	Implied shipments (k)	HBM chip density (GB)	HBM chip units	Total HBM size (GB)	HBM generation	HBM vendor	Total HBM demand (k GB)
AI GPU (2025e)										
NVIDIA	H200/H20	50	27	1,350	24	6	141	HBM3e	MU/Samsung	190,350
	B200	160	14	2,240	24	8	192	HBM3e	Hynix/Micron/Samsung	430,080
	B300A	28	27	745	24	4	96	HBM3e	Hynix/Micron	71,539
	B300	202	14	2,834	36	8	288	HBM3e 12hi	Hynix/Micron/Samsung	816,077
AMD	MI300	5	12	60	24	8	192	HBM3	Samsung	11,520
	MI325	20	12	240	36	8	288	HBM3e 12hi	Samsung	69,120
	MI355	25	12	300	36	8	288	HBM3e 12hi	Samsung	86,400
Habana	Gaudi 3	9	14	126	16	8	128	HBM2e	Samsung	16,128
AI ASIC (2025e)										
Google	TPU v5 training (v5p)	10	25	250	16	6	96	HBM2e	Samsung	24,000
	TPU v6 inferencing (v6e)	60	30	1,800	16	4	64	HBM3	Samsung/Micron	115,200
	TPU v6 training (v6p)	15	20	300	24	8	192	HBM3	Hynix/Samsung/Micron	57,600
AWS	Trainium 2	40	16	640	24	4	96	HBM3e	Samsung	61,440
	Trainium 2.5	25	16	400	36	4	144	HBM3e 12hi	Samsung	57,600
	Trainium 3	5	17	85	36	8	288	HBM3e 12hi	Hynix/Samsung/Micron	24,480
Tesla	Dojo 1	NA	NA	200	32	5	160	HBM2e/HBM3	Samsung	1,280
	FSD	NA	NA	1,600	NA	NA	NA	NA	NA	-
Microsoft	Maia 200	2	29	58	16	4	64	HBM3	Samsung	3,712
	Maia 200 Enhancement	1	29	29	16	4	64	HBM3	Samsung	1,856
Chinese GPU/ASIC	Chinese GPU/ASIC	5.5	20	110	16	4	64	NA	Samsung	7,040
Total		663								2,045,422
Total HBM demand (mn Gb)										16,363

Source: Company data, Morgan Stanley Research (e) estimates; note: estimates are compiled using our supply chain checks

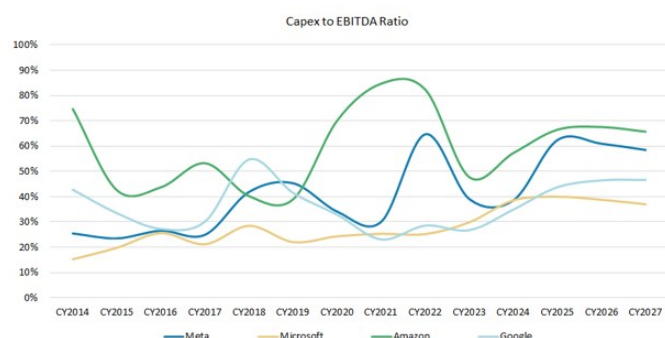
Global AI capex update

We think cash flow analysis could support our expectations of an ongoing rise in capex from large cloud service providers. Morgan Stanley forecasts that the top four US hyperscalers will generate US\$550bn in operating cash flow in 2025. Thus, we believe these companies have the ability to keep investing in data centers for AI purposes.

We also look at depreciation as a percentage of total expenses. In past years, given the heavy investment in data centers, depreciation on data center customers' total expenses rose substantially (from 3-7% in 2012 to 5-10% in 2023). We expect depreciation to continue to grow amid rising capex from the top four US hyperscalers, up to 10-14% in 2025e.

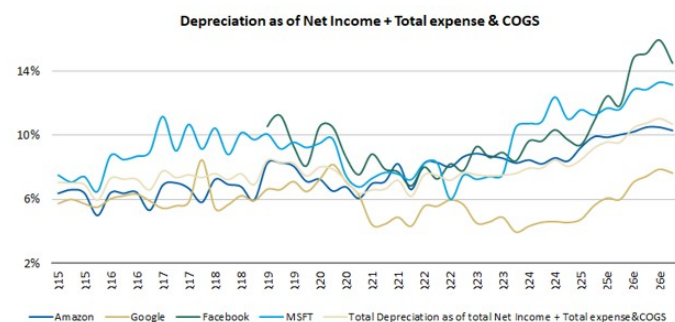
We also see average AI capex/EBITDA at around 50% in 2025, while we think hyperscalers' pockets are still deep enough for further spending.

Exhibit 19: Capex to EBITDA ratio should remain stable at around 50%



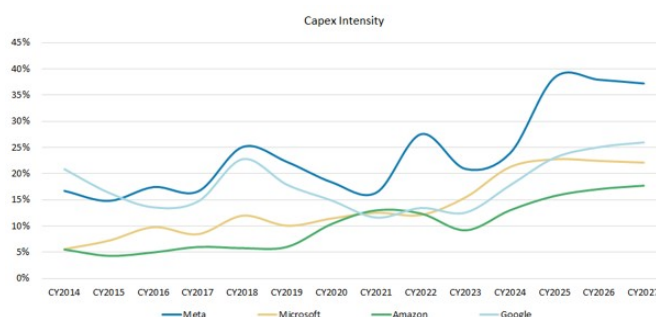
Source: Company data, Morgan Stanley Research (E) estimates

Exhibit 21: Depreciation has become a bigger portion of expenses in recent years



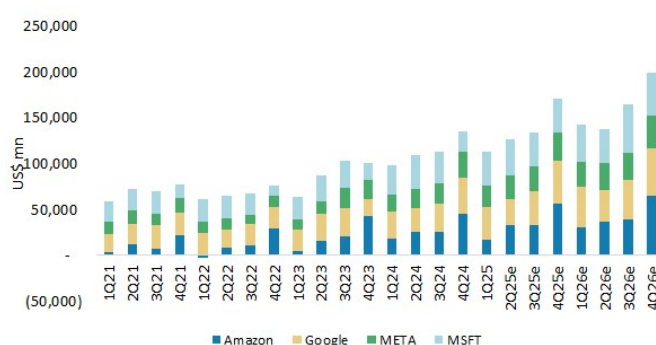
Source: Company data, Morgan Stanley Research (e) estimates

Exhibit 20: Capex intensity remains stable among top four US hyperscalers



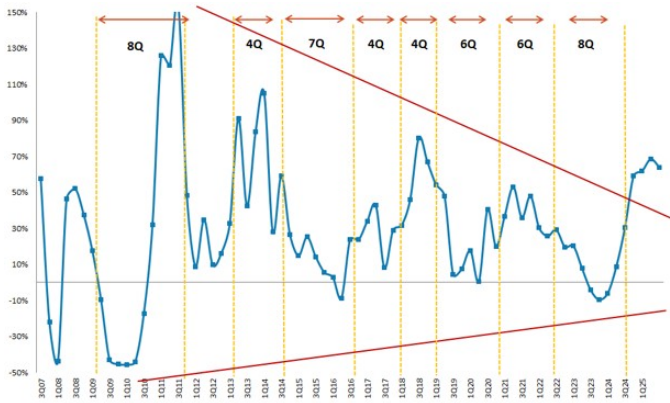
Source: Company data, Morgan Stanley Research (E) estimates

Exhibit 22: Rising operating cash flow could continue to support growth in hyperscalers' capex



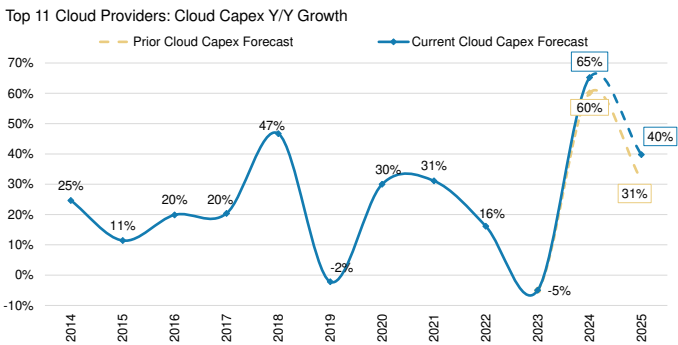
Source: Company data, Morgan Stanley Research (e) estimates

Exhibit 23: Top four CSPs (Amazon, Google, Microsoft and Meta) capex +64% Y/Y in 1Q25



Source: Company data, Morgan Stanley Research (e) estimates

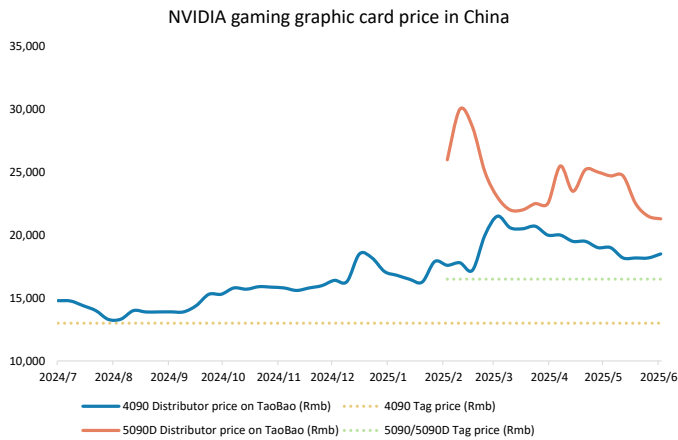
Exhibit 24: Following Amazon's 1Q25 earnings, our cloud capex tracker now points to 40% Y/Y growth in 2025, up from the prior forecast of +31% Y/Y



Source: Company data, FactSet, Morgan Stanley Research estimates, irs.gov. Note: Cloud capex includes capex from Alphabet, Amazon, Microsoft, Meta Platforms, Tencent, Baidu, Apple, IBM, and Oracle. Forward estimates include MSe for Tencent and Baidu, consensus est. for the others; Alphabet's CY25 capex is based on the company's guidance, while the remaining CY25 numbers are Consensus estimates.

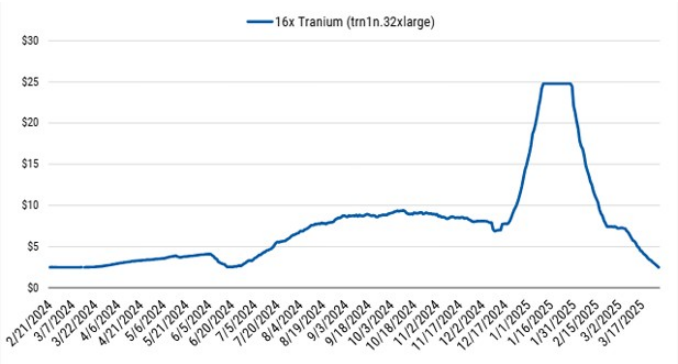
AI GPU and ASIC rental price tracker

Exhibit 25: Retail price of NVIDIA 4090 and 5090 graphic card is slightly decreasing although China's AI inference demand is still very strong



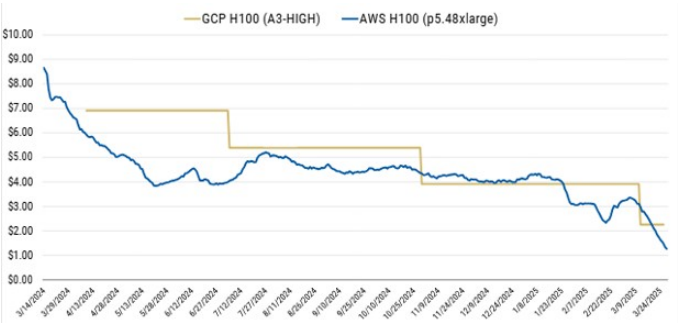
Source: TaoBao, Morgan Stanley Research

Exhibit 27: AI ASIC equivalent computing power - 16x Inferentia 2 per hour



Source: Company data, Morgan Stanley Research

Exhibit 26: AI GPU H100 per GPU per hour as of the end of March

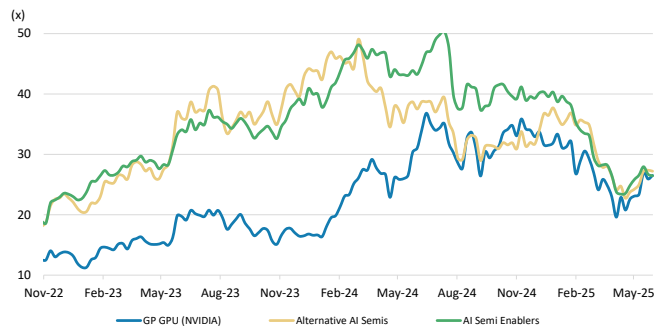


Source: Company data, Morgan Stanley Research

AI semis – P/E multiples, revenue exposure, sales tracker

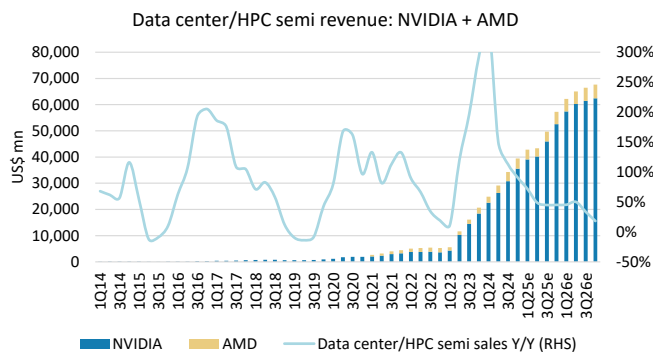
Major AI GPU vendors – NVIDIA sales and inventory trends

Exhibit 28: P/E multiple trend of AI semis



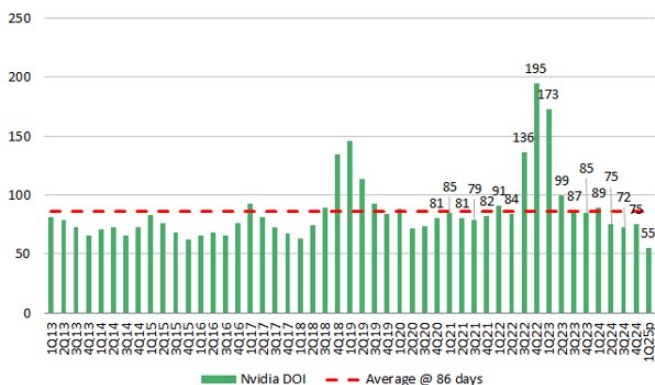
Alternative AI semis group: AMD, Alchip, GUC, Andes, Marvell, Broadcom. AI semi enablers group: TSMC, Synopsys, Cadence, ASML, BESi, Ibiden, KYEC, Advantest.
Source: Company data, Morgan Stanley Research.

Exhibit 30: We continue to see AI chip quarterly revenue increasing



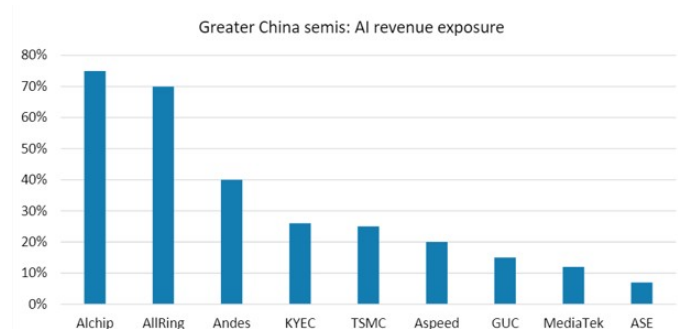
Source: Company data, Refinitiv, Morgan Stanley Research (e) estimates

Exhibit 32: NVIDIA's inventory days/level



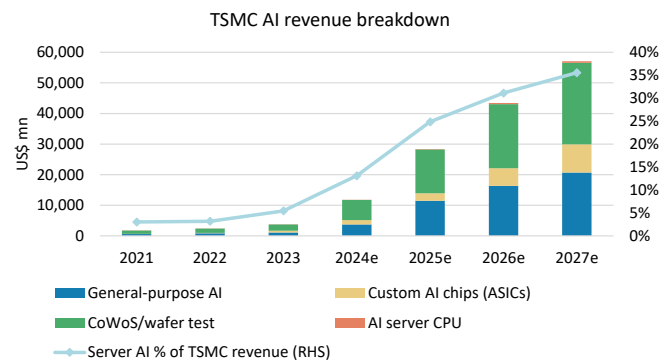
Source: Company data, Refinitiv, Morgan Stanley Research

Exhibit 29: Greater China AI semi revenue exposure as of 2025e



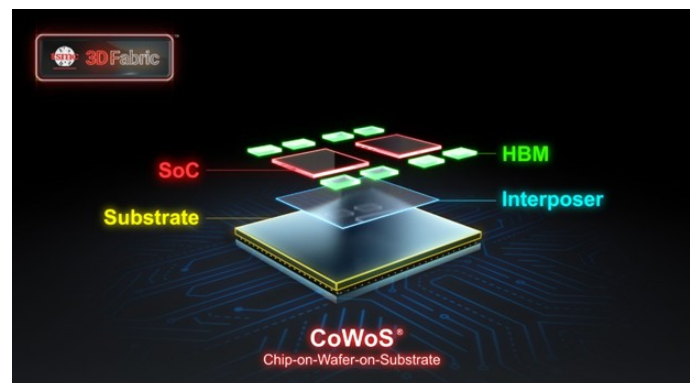
Source: Morgan Stanley Research (e) estimates

Exhibit 31: AI accounts for a mid-teens share of TSMC's 2024 total revenue and 25% of TSMC's 2025e total revenue



Source: Company data, Morgan Stanley Research (e) estimates

Exhibit 33: What is CoWoS? Chip on Wafer (interposer) on Substrate



Source: TSMC

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AI supply chain trackers

Asia Technology: AI Supply Chain: Blackwell Orders Are Intact, while AI PC Launch May Be Delayed (15 May 2025)

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Alchip Technologies Ltd: Reaffirming 3nm project win and its on-track schedule; OW (2 Jun 2025)

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Underweight/Sell	602	16%	74	9%	12%	235	14%
Total	3,749		825			1717	

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INDUSTRY COVERAGE: Greater China Technology Semiconductors

COMPANY (TICKER)	RATING (AS OF)	PRICE* (06/02/2025)
Charlie Chan		
ACM Research Inc (ACMR.O)	O (03/07/2023)	US\$22.55
Advanced Micro-Fabrication Equipment Inc (688012.SS)	O (11/06/2023)	Rmb175.01
Advanced Wireless Semiconductor Co (8086.TWO)	E (10/24/2024)	NT\$74.10
Alchip Technologies Ltd (3661.TW)	O (05/14/2021)	NT\$2,720.00
Andes Technology Corp (6533.TW)	O (08/04/2022)	NT\$256.50
ASE Technology Holding Co. Ltd. (3711.TW)	O (09/15/2024)	NT\$134.50
Global Unichip Corp (3443.TW)	O (07/27/2024)	NT\$1,095.00
GlobalWafers Co Ltd (6488.TWO)	E (02/22/2023)	NT\$293.00
Gudeng Precision (3680.TWO)	O (08/27/2024)	NT\$350.00
Hua Hong Semiconductor Ltd (1347.HK)	E (05/09/2025)	HK\$30.75
King Yuan Electronics Co Ltd (2449.TW)	O (03/03/2023)	NT\$92.50
M31 Technology Corp (6643.TWO)	E (09/23/2024)	NT\$502.00
Maxscend Microelectronics Co Ltd (300782.SZ)	U (01/11/2021)	Rmb70.46
MediaTek (2454.TW)	O (09/12/2023)	NT\$1,260.00

Nanya Technology Corp. (2408.TW)	E (05/15/2025)	NT\$45.80
NAURA Technology Group Co Ltd (002371.SZ)	O (11/06/2023)	Rmb418.56
Phison Electronics Corp (8299.TWO)	O (03/11/2025)	NT\$497.50
SG Micro Corp. (300661.SZ)	O (06/17/2024)	Rmb92.22
Silergy Corp. (6415.TW)	O (02/13/2025)	NT\$344.00
SMIC (0981.HK)	E (04/09/2025)	HK\$40.10
TSMC (2330.TW)	O (02/07/2022)	NT\$946.00
UMC (2303.TW)	E (10/28/2024)	NT\$46.95
Vanguard International Semiconductor (5347.TWO)	U (08/05/2024)	NT\$80.60
Will Semiconductor Co Ltd Shanghai (603501.SS)	O (05/09/2025)	Rmb124.60
WIN Semiconductors Corp (3105.TWO)	E (05/12/2025)	NT\$83.20
Daisy Dai, CFA		
ASMP Ltd (0522.HK)	E (11/20/2024)	HK\$53.55
China Resources Microelectronics Limited (688396.SS)	U (09/02/2024)	Rmb47.63
Elan Microelectronics Corp (2458.TW)	E (04/08/2025)	NT\$134.00
Empyrean Technology Co Ltd (301269.SZ)	E (01/17/2025)	Rmb127.07
Hangzhou Silan Microelectronics Co. Ltd. (600460.SS)	E (04/09/2024)	Rmb24.02
JCET Group Co Ltd (600584.SS)	U (09/25/2024)	Rmb32.16
Shanghai Anlogic Infotech Co Ltd (688107.SS)	E (05/14/2024)	Rmb26.58
Shanghai Fudan Microelectronics (1385.HK)	O (03/07/2025)	HK\$28.80
SICC Co Ltd (688234.SS)	O (09/18/2023)	Rmb60.00
StarPower Semiconductor Ltd (603290.SS)	O (03/01/2022)	Rmb80.20
Unigroup Guoxin Microelectronics Co Ltd (002049.SZ)	U (01/10/2023)	Rmb63.97
Universal Scientific Ind. (Shanghai) (601231.SS)	E (10/23/2024)	Rmb13.52
Yangjie Technology (300373.SZ)	O (06/10/2022)	Rmb47.45
Daniel Yen, CFA		
AP Memory Technology Corp (6531.TW)	E (09/15/2024)	NT\$272.50
ASMedia Technology Inc (5269.TW)	E (04/08/2025)	NT\$1,895.00
Aspeed Technology (5274.TWO)	E (09/15/2024)	NT\$3,945.00
Egis Technology Inc (6462.TWO)	E (03/26/2024)	NT\$93.10
Espressif Systems (688018.SS)	O (05/15/2023)	Rmb184.13
GigaDevice Semiconductor Beijing Inc (603986.SS)	O (05/15/2025)	Rmb112.30
Macronix International Co Ltd (2337.TW)	U (10/19/2021)	NT\$21.60
Montage Technology Co Ltd (688008.SS)	O (11/08/2023)	Rmb76.32
Novatek (3034.TW)	O (03/18/2024)	NT\$492.50
Nuvoton Technology Corporation (4919.TW)	O (10/29/2024)	NT\$69.80
Parade Technologies Ltd (4966.TWO)	U (06/04/2024)	NT\$547.00
Powerchip Semiconductor Manufacturing Co (6770.TW)	E (06/13/2024)	NT\$14.75
Realtek Semiconductor (2379.TW)	U (09/15/2024)	NT\$525.00
Shenzhen Goodix Technology Co Ltd (603160.SS)	E (06/12/2024)	Rmb68.98
Sino Wealth Electronic (300327.SZ)	U (03/31/2025)	Rmb20.69
Winbond Electronics Corp (2344.TW)	O (03/11/2025)	NT\$16.90
WPG Holdings (3702.TW)	E (11/16/2023)	NT\$65.20
Duan Liu		
Dosilicon Co Ltd (688110.SS)	U (09/06/2024)	Rmb31.28
Shenzhen Longsys Electronics Co Ltd (301308.SZ)	O (03/11/2025)	Rmb71.18
Tiffany Yeh		
AllRing Tech Co. (6187.TWO)	E (05/13/2025)	NT\$304.00
FOCI Fiber Optic Communications Inc (3363.TWO)	O (01/15/2025)	NT\$243.00
Himax Technologies Inc (HIMX.O)	O (05/09/2025)	US\$8.15
Silicon Motion (SIMO.O)	O (05/06/2024)	US\$61.20

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* Historical prices are not split adjusted.

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